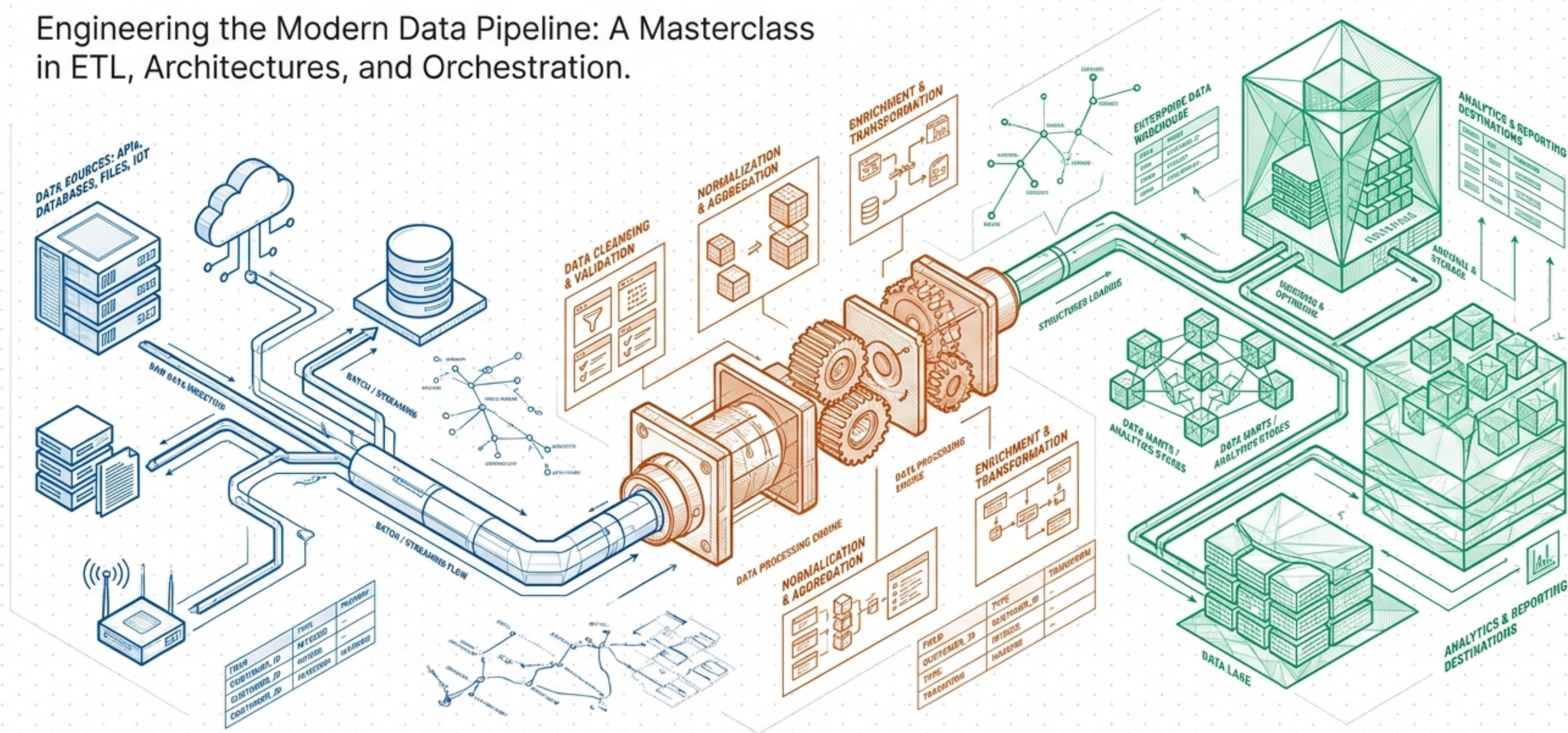
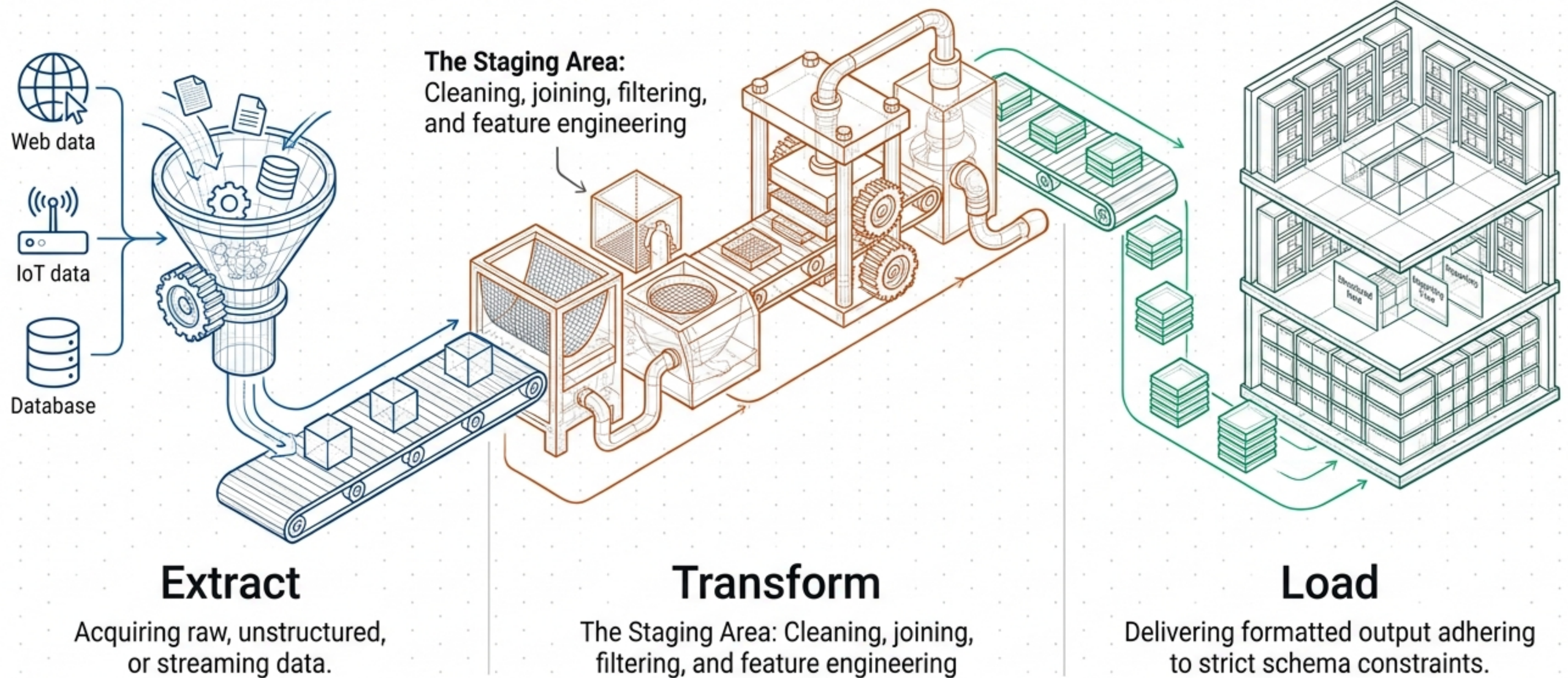


THE DATA REFINERY BLUEPRINT

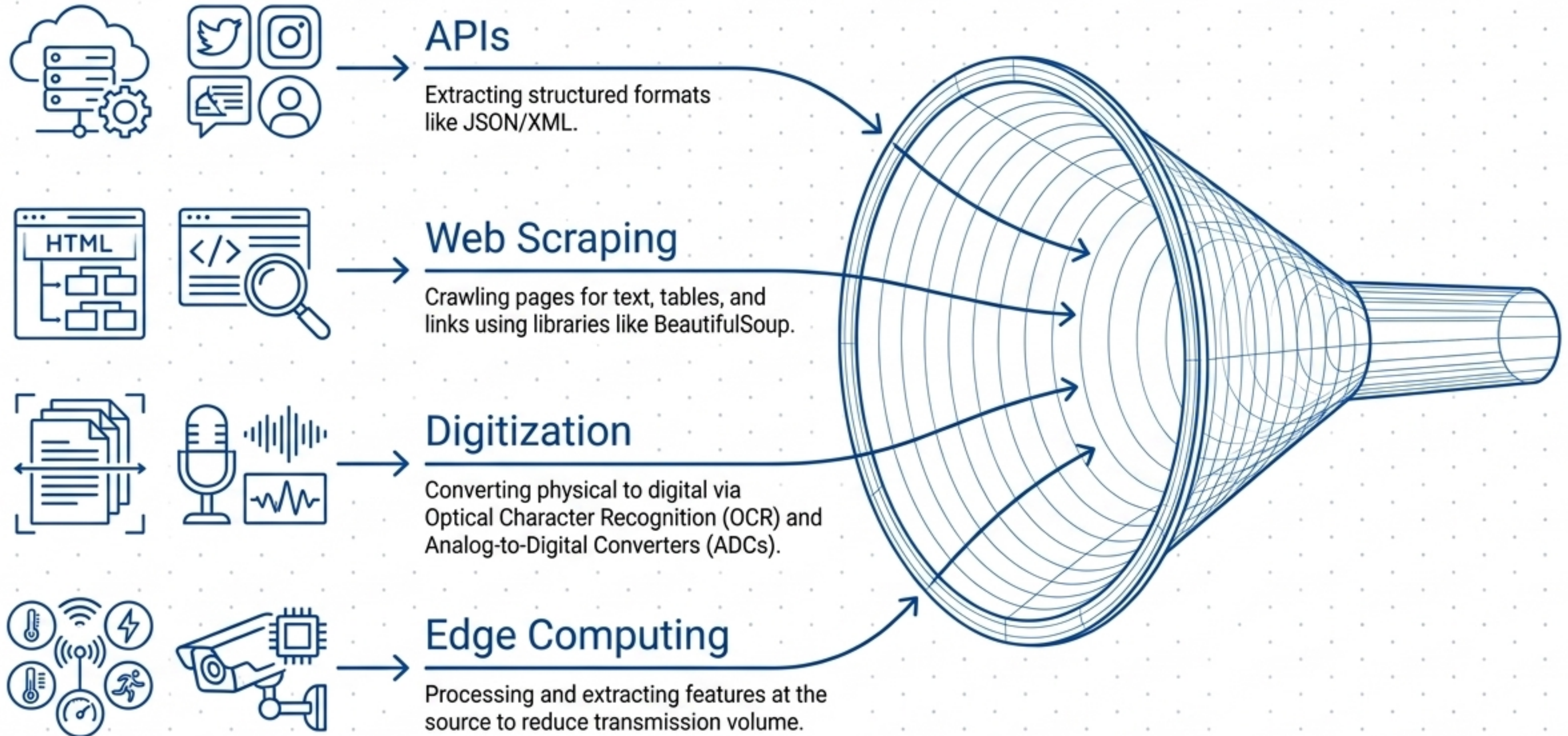
Engineering the Modern Data Pipeline: A Masterclass in ETL, Architectures, and Orchestration.



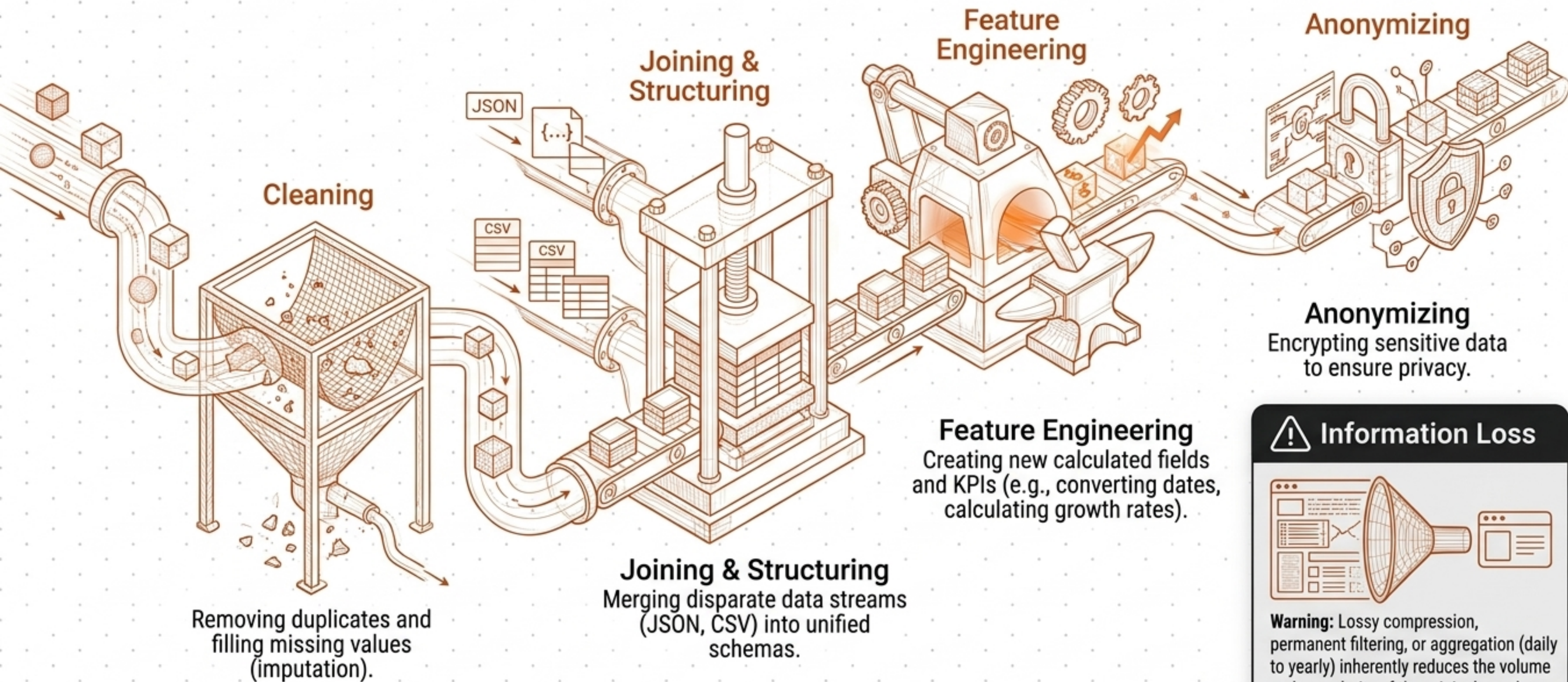
The Core Engine: Deconstructing the ETL Pipeline



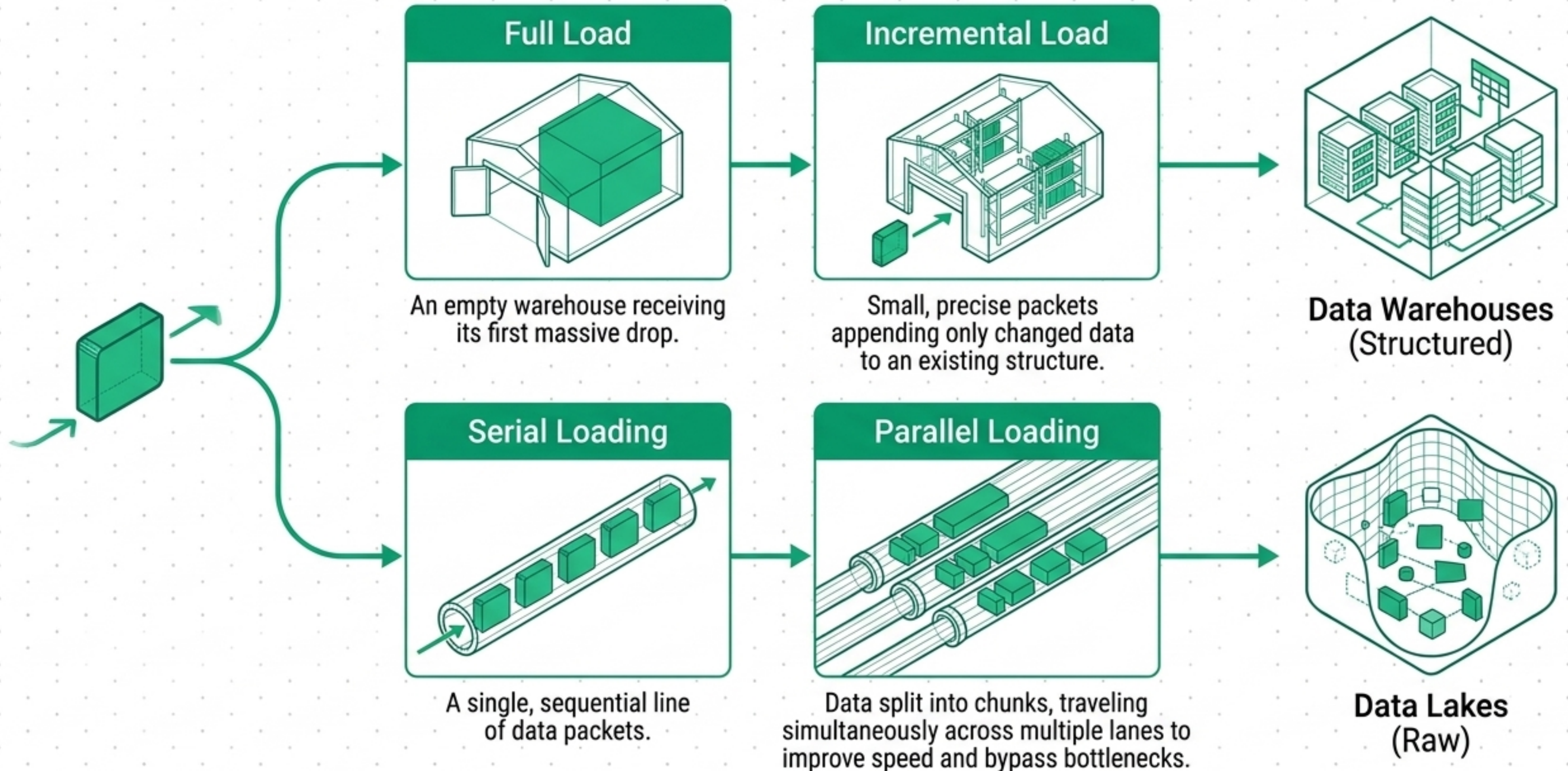
Phase 1: Sourcing the Raw Material



Phase 2: The Staging Area and Data Wrangling

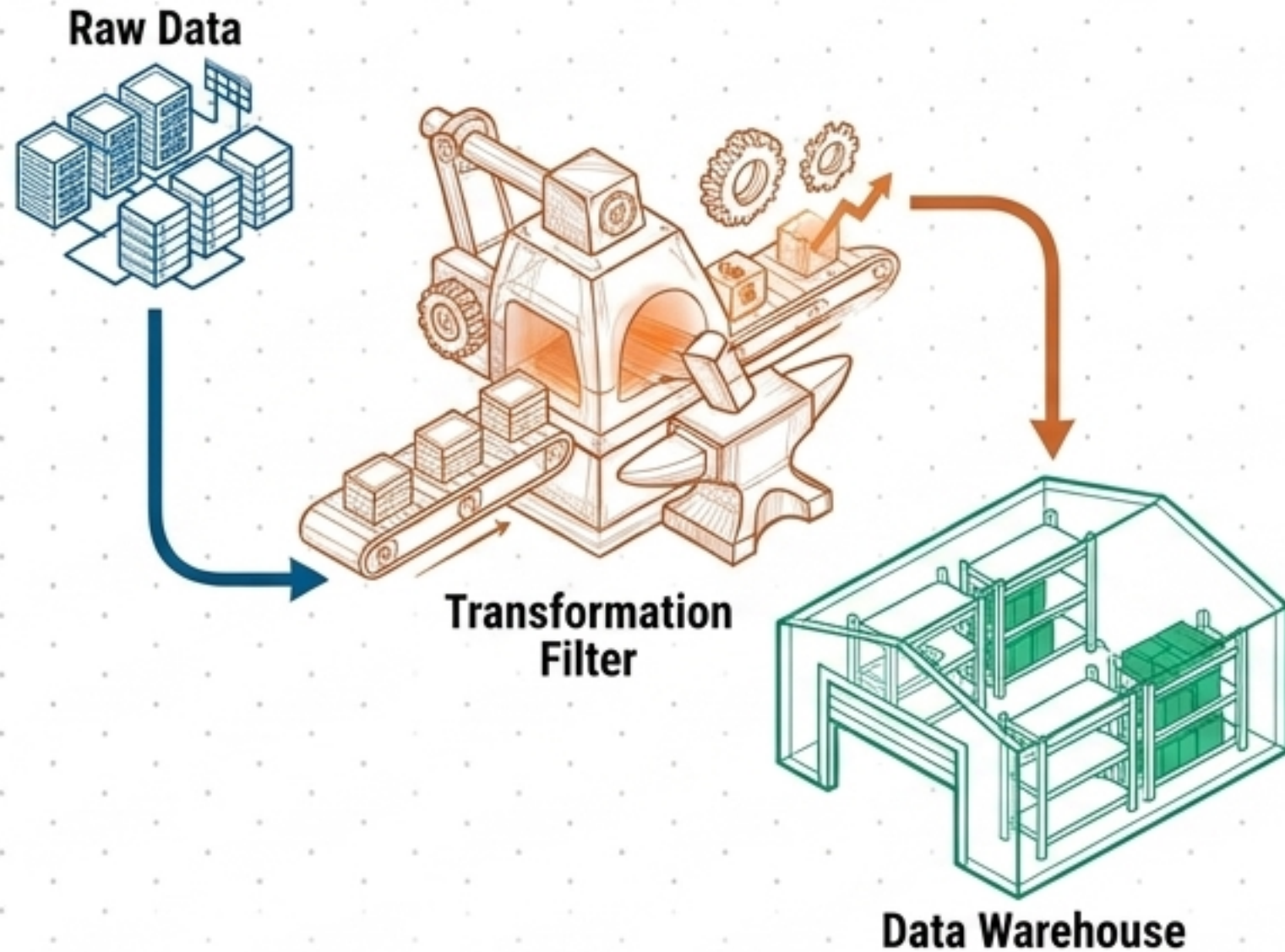


Phase 3: Delivering the Refined Output



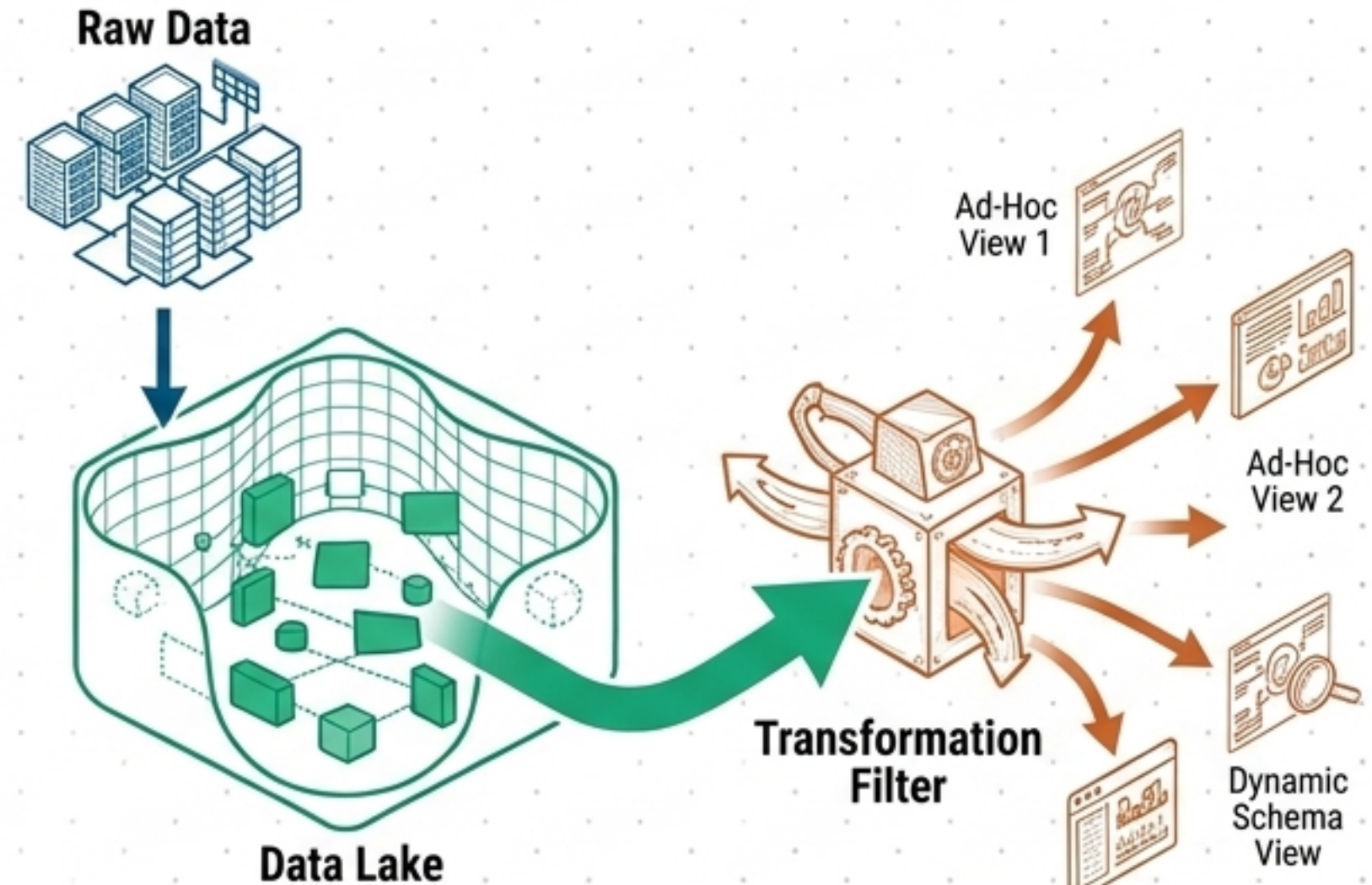
The Paradigm Shift: Schema-on-Write (ETL) vs. Schema-on-Read (ELT)

ETL / Schema-on-Write



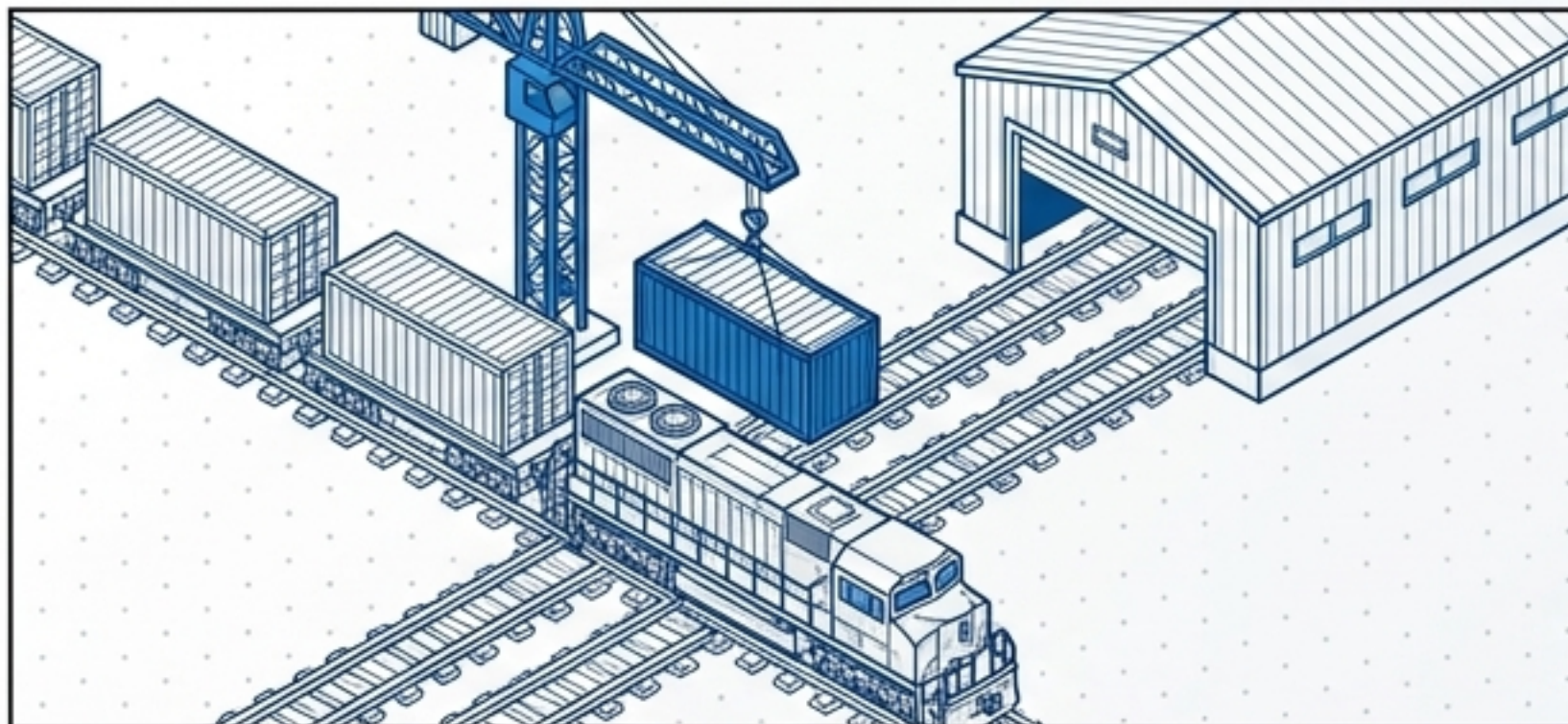
- Highly structured, consistent, rigorous upfront preprocessing.
- Ideal for stable and fast queries.

ELT / Schema-on-Read



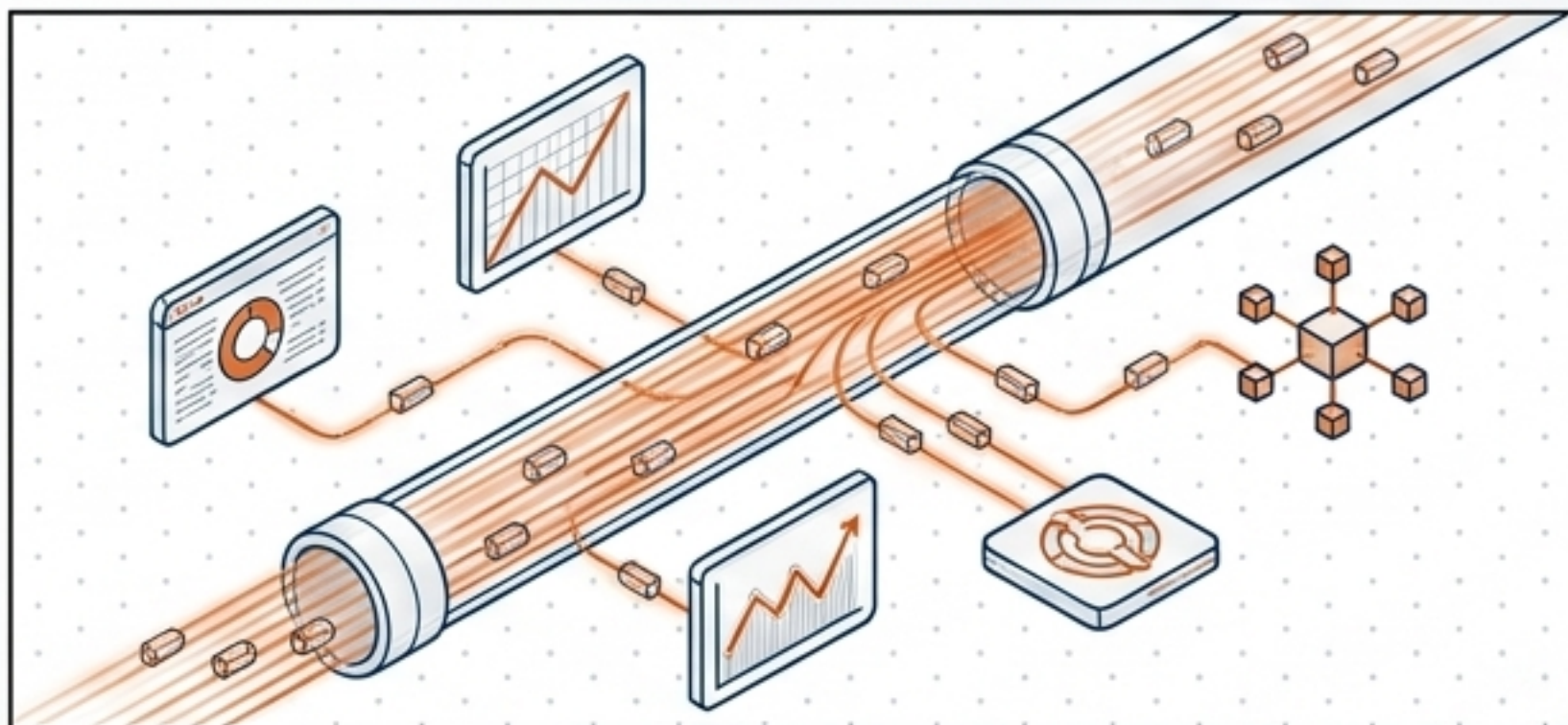
- High versatility, flexible, raw data preservation.
- Supports multiple ad-hoc views generated upon query.

The Velocity Spectrum: Batch vs. Streaming Execution



Batch Pipeline

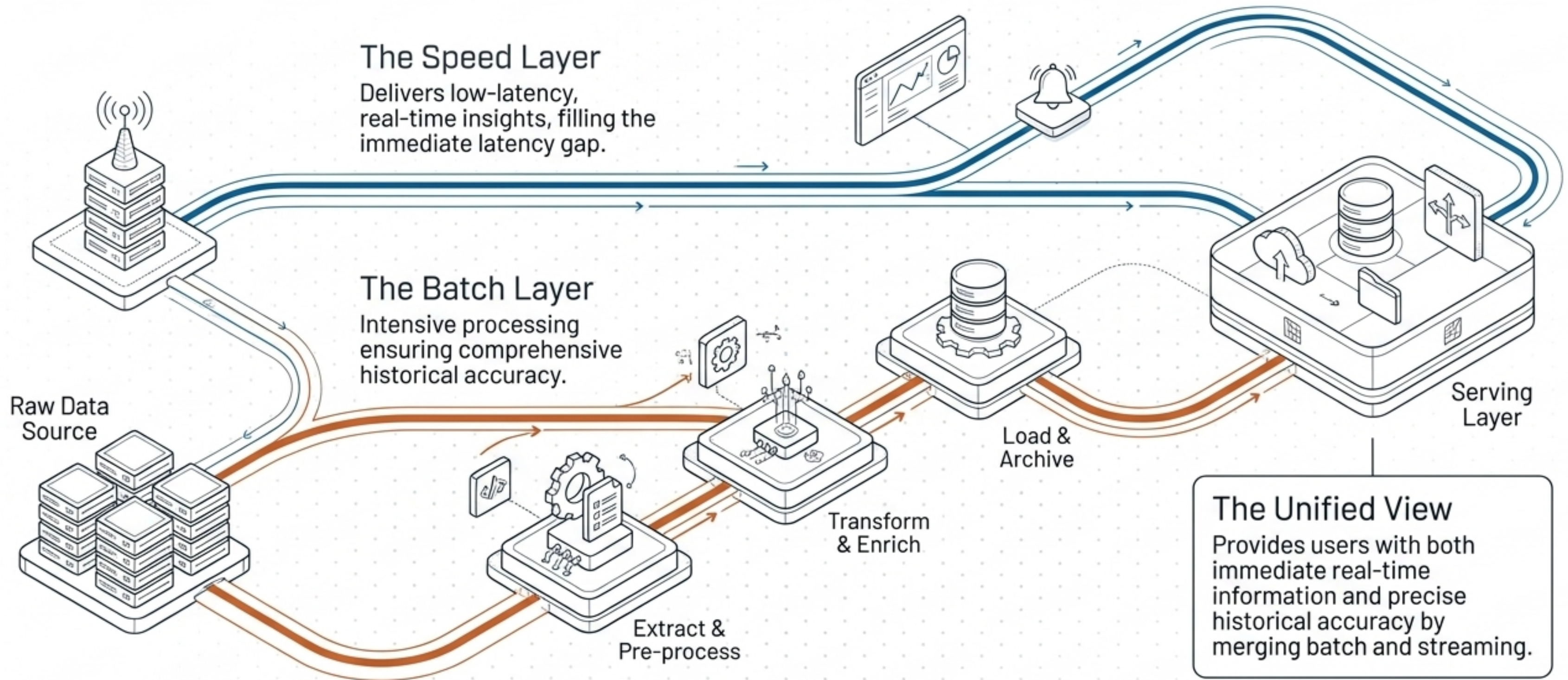
Trigger:	Time-based (e.g., Cron job, hourly/daily)
Method:	Pull (warehouse requests data)
Use Cases:	Sales forecasting, billing, periodic backups



Streaming Pipeline

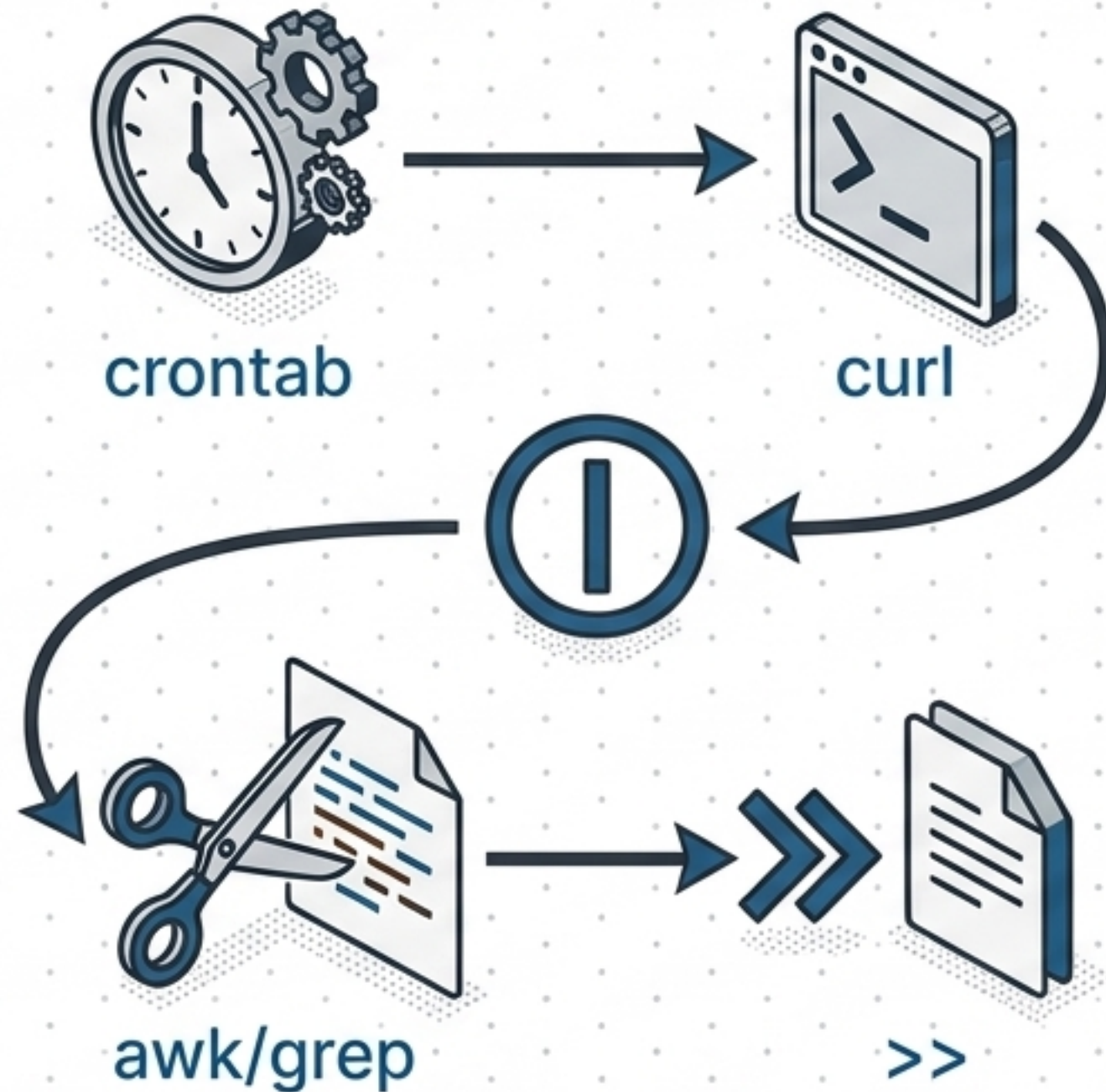
Trigger:	Event-driven (continuous)
Method:	Push (source sends data as it arrives)
Use Cases:	Fraud detection, stock trading, social media feeds

Bridging the Gap: The Lambda Architecture Blueprint



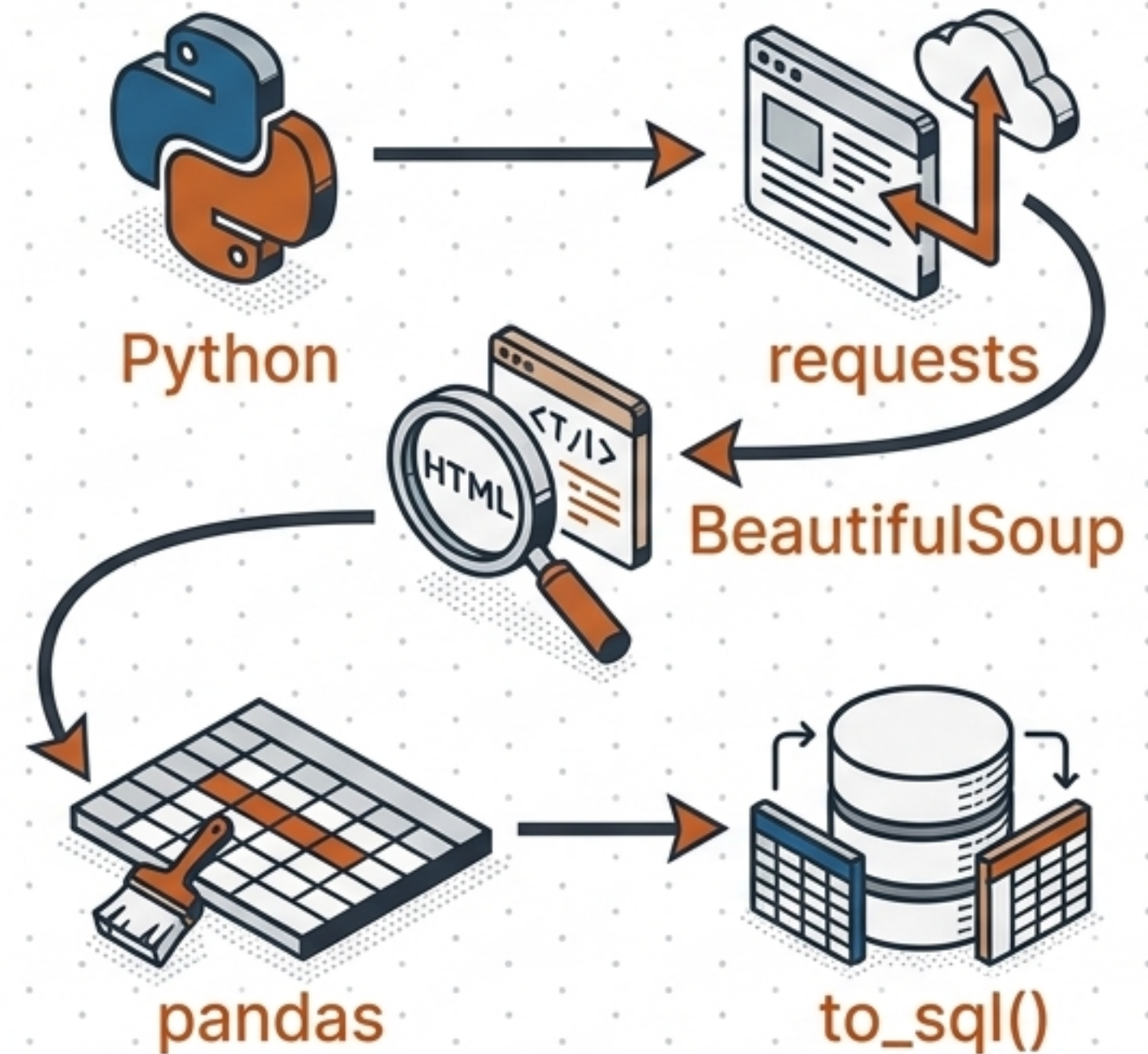
The Builder's Toolkit: Scripting the Pipeline

Bash Scripting



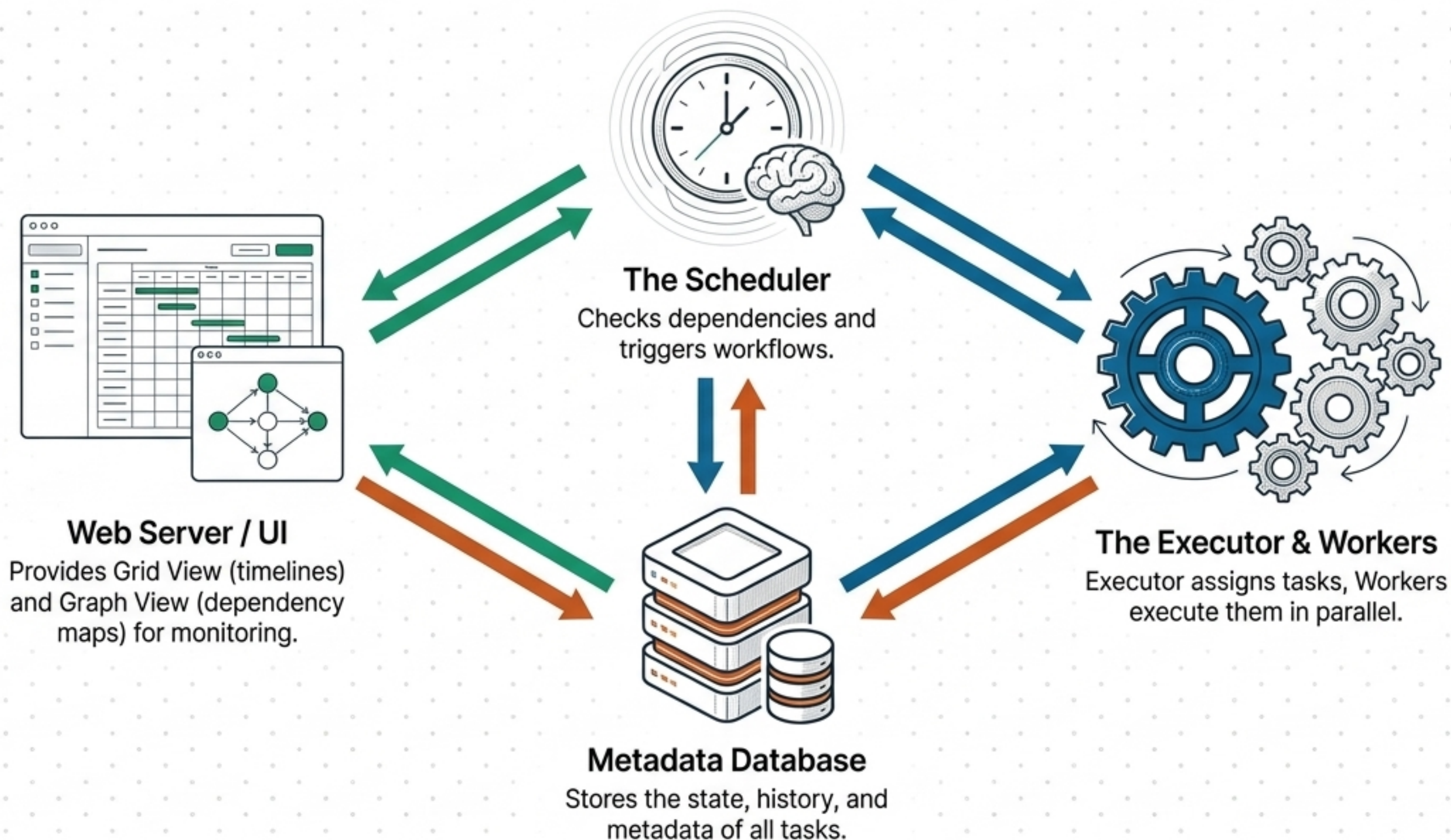
Precise data extraction & filtering chain.

Python Scripting

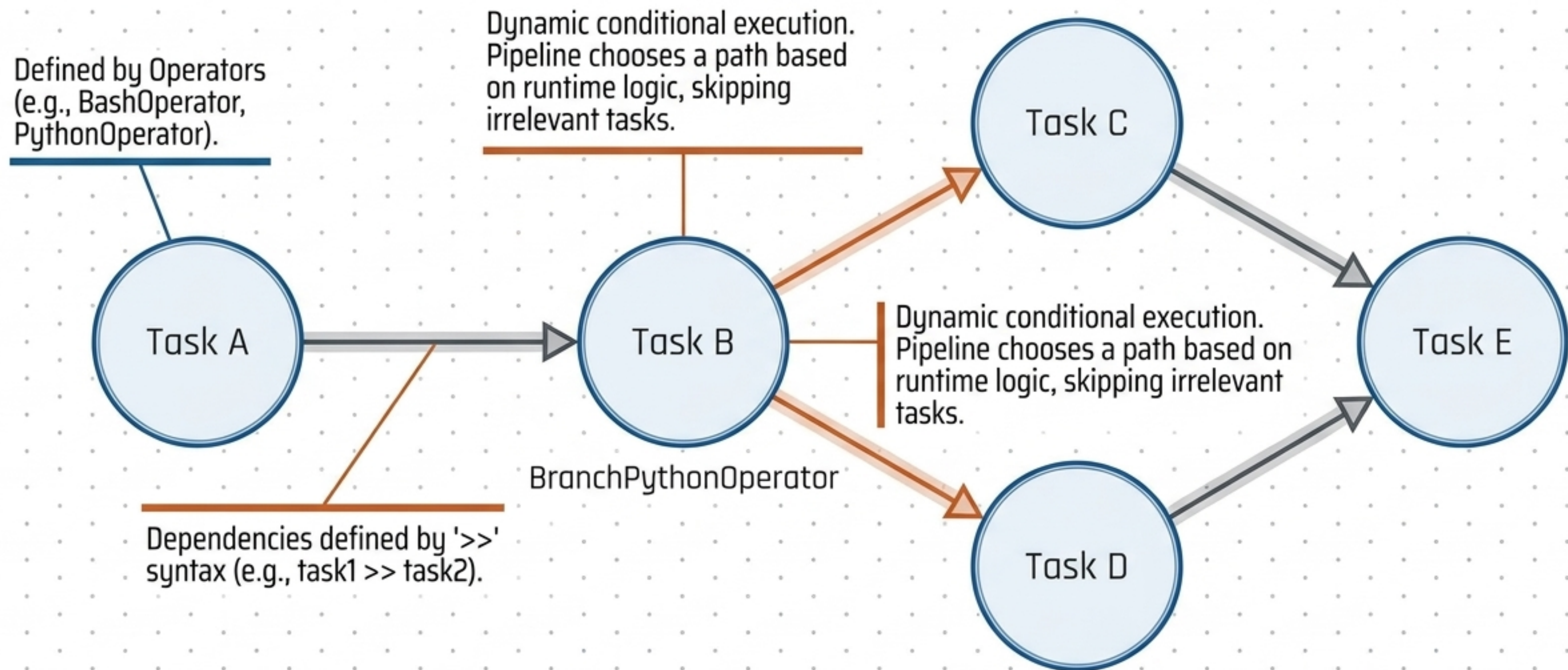


Advanced parsing, cleaning & loading workflow.

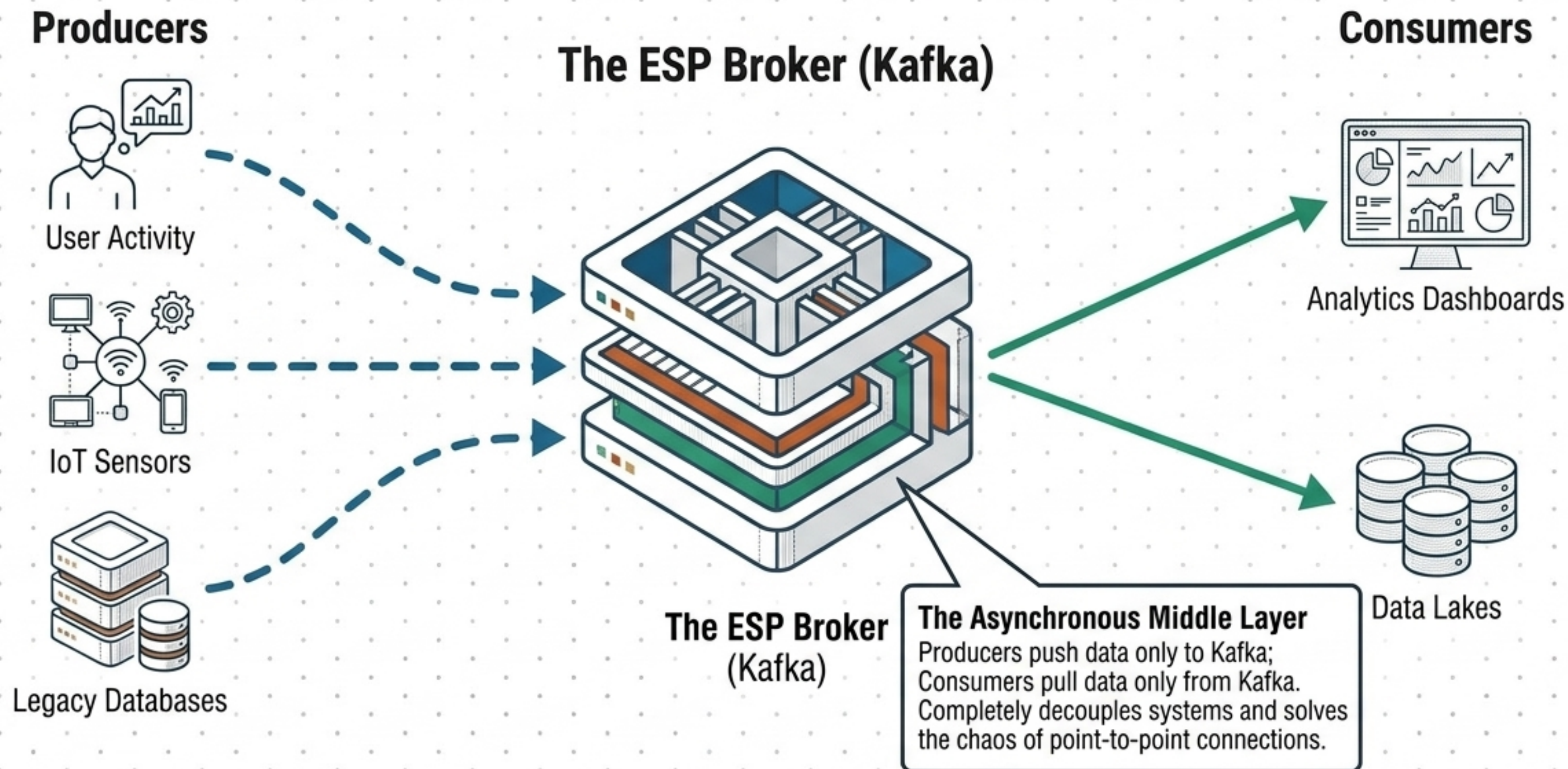
Orchestration at Scale: Inside Apache Airflow



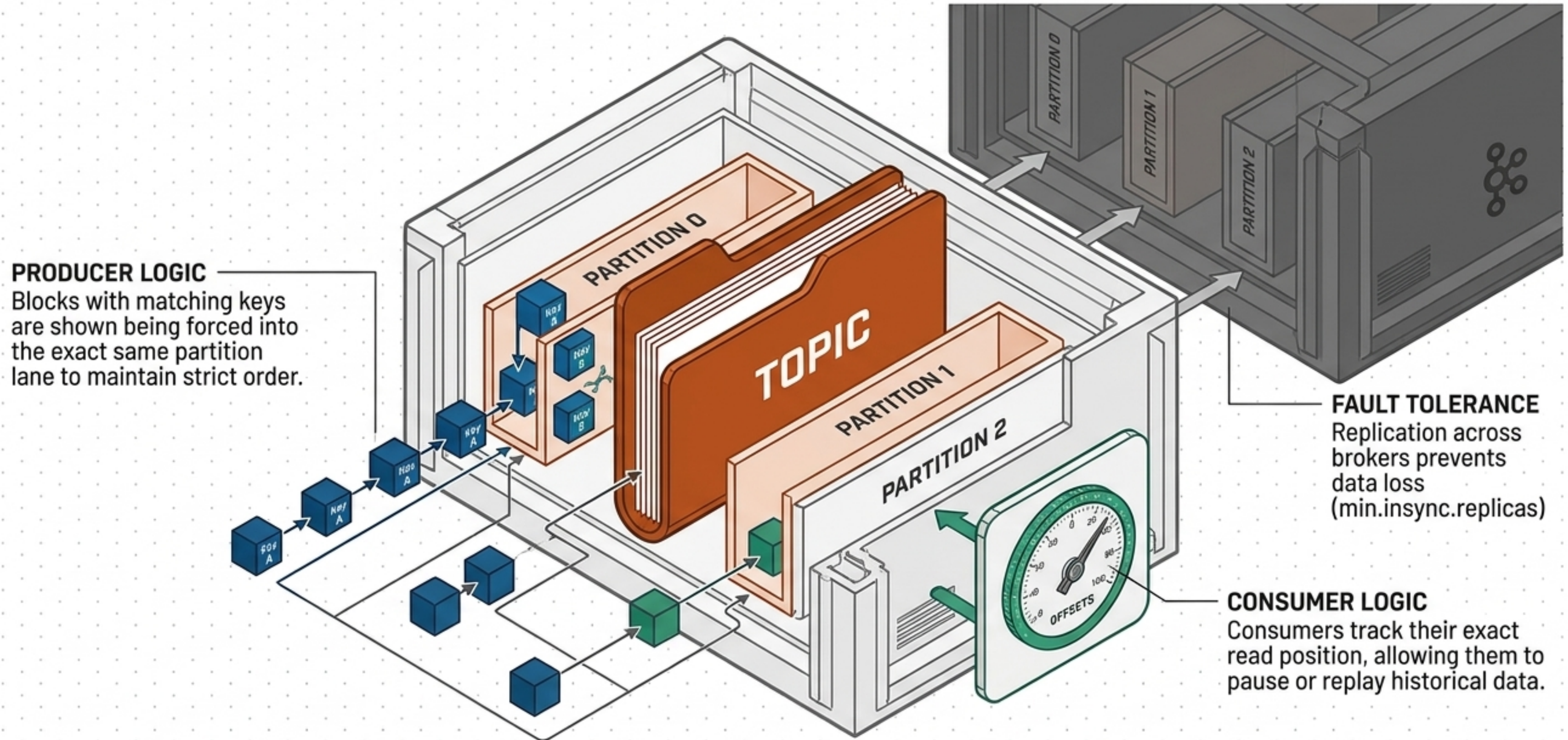
Decoding the Workflow: Anatomy of a DAG



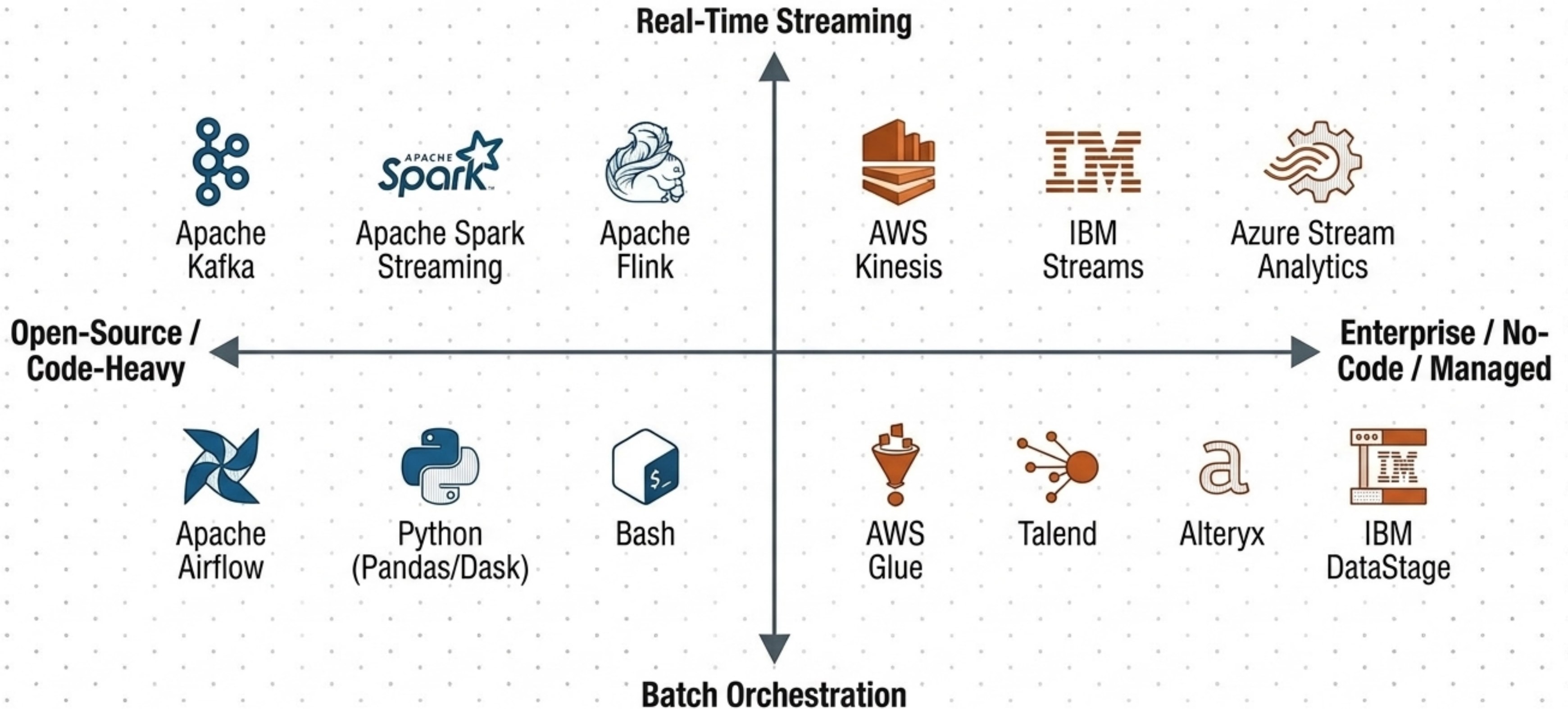
The Streaming Paradigm: Apache Kafka as the Central Nervous System



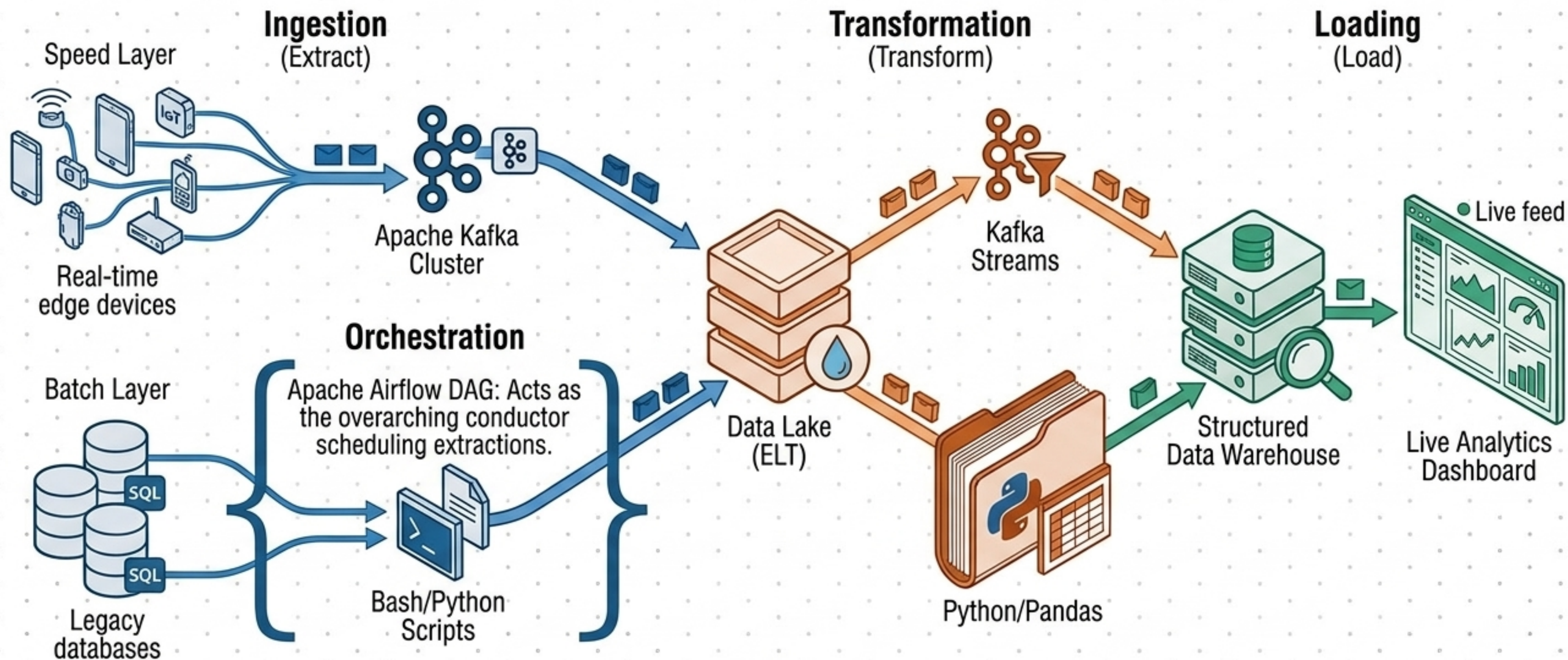
Inside the Broker: Topics, Partitions, and Fault Tolerance



The Modern Tool Landscape



Synthesis: The Modern Enterprise Pipeline



True mastery is not in a single tool, but in engineering the harmonious flow between them.